ModelIt!

Teacher's Guide

How Does Your Immune System Learn?

Modeling Vaccine Mechanisms and Herd Immunity Thresholds

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-LS1-2, HS-LS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2, HS-LS1-4

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms. Use a model to illustrate the role of cellular division (mitosis) in producing and maintaining complex organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to illustrate how the adaptive immune system produces immunological memory and how population-level vaccination coverage creates herd immunity.
Disciplinary Core Idea	**LS1.A Structure and Function** — The immune system's hierarchical organization — from molecular antigen recognition through cellular activation to systemic memory — provides a specific defense function through coordinated interactions among specialized cells.
Crosscutting Concept	**Systems and System Models** — Students model immunity at two scales: the cellular scale (individual immune response) and the population scale (herd immunity dynamics), understanding how individual-level processes create emergent population-level protection.

CAST Testing Alignment

MULTIPLE CHOICE:

A disease has an R0 of 6, meaning each infected person spreads it to 6 others. Using the herd immunity formula (1 - 1/R0), what percentage of the population must be immune to stop transmission? A) 50% B) 67% C) 83% D) 93%

CONSTRUCTED RESPONSE:

Using your model, explain why a person who was vaccinated 10 years ago is still protected even though their antibody levels have declined significantly. Describe the role of memory cells and how the secondary immune response differs from the primary response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The development of vaccines is arguably the greatest achievement in the history of medicine — vaccines have saved an estimated 154 million lives over the past 50 years. Yet many people fundamentally misunderstand how vaccines work, confusing antibody levels with protection and individual immunity with population immunity. A deep understanding of the adaptive immune system and herd immunity dynamics is essential for making informed personal and public health decisions.

The Adaptive Immune System: Learning Through Molecular Recognition

The adaptive immune system relies on two types of lymphocytes: B cells (which produce antibodies) and T cells (which kill infected cells and coordinate the immune response). Each B cell and T cell carries unique receptors generated through random gene rearrangement — your body produces billions of different receptor variants, ensuring that at least a few will recognize virtually any pathogen. When a pathogen's antigens bind to matching receptors, those specific B and T cells are activated, multiply rapidly (clonal expansion), and mount a targeted attack. This process takes 7-14 days for a first exposure — which is why you get sick before your immune system can clear the infection.

How Vaccines Create Memory Without Disease

Vaccines introduce antigens in a form that triggers the adaptive immune response without causing illness. Traditional vaccines use weakened (attenuated) or killed pathogens. Modern approaches include subunit vaccines (purified protein fragments), mRNA vaccines (genetic instructions for cells to make the antigen), and viral vector vaccines (harmless viruses carrying pathogen genes). All approaches accomplish the same goal: activating pathogen-specific B cells and T cells, driving clonal expansion, and producing memory cells — all without the pathogen ever establishing an infection.

Memory Cells: The Permanent Guard

After the initial immune response clears the antigen, 90-95% of activated B and T cells die through apoptosis. But a critical fraction — memory B cells and memory T cells — survive and persist for years or decades in lymph nodes, bone marrow, and tissues throughout the body. These memory cells retain the exact receptor specificity for the pathogen. Upon re-exposure (whether from infection or a booster vaccine), memory cells activate within hours, producing a secondary response that is 10-100 times faster and stronger than the primary response. This is why vaccinated people either don't get sick at all or experience much milder illness.

Herd Immunity: Individual Decisions, Collective Consequences

Herd immunity is the population-level protection that emerges when enough individuals are immune to break transmission chains. The threshold is calculated as 1 - (1/R0): measles (R0≈15) requires approximately 93% immunity; pertussis (R0≈12) requires 92%; COVID-19 original strain (R0≈3) required 67%. Below the threshold, outbreaks can occur because susceptible individuals are connected in the population. Above the threshold, even susceptible individuals (infants too young to vaccinate, immunocompromised patients) are protected because the pathogen cannot find enough hosts to sustain spread. Every person who chooses not to vaccinate shifts the population closer to — or below — this critical threshold.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify vaccination rate, antigen exposure, antibody production, memory cell population, herd immunity level, and disease transmission rate as the key components of the vaccination-immunity system.

E **Establish Relationships**

Students determine that vaccination rate and antigen exposure are external inputs controlled by public health policy and vaccine design, while antibody production, memory cells, herd immunity, and disease transmission are biological and epidemiological outcomes that emerge from the interaction of individual immune responses at the population level.

V **Visualize & Model**

Students build a computational model operating at two scales: individual immune response (antigen to memory cells) and population dynamics (vaccination rate to herd immunity threshold).

E **Evaluate Outcomes**

Students run individual immune response, herd immunity buildup, and under-vaccinated outbreak scenarios to understand both the cellular basis of protection and the population-level dynamics of disease control.

R **Revise & Extend**

Students add vaccine efficacy rate, variant evolution, or vaccine hesitancy to explore real-world complications that affect both individual and population immunity.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You got vaccinated as a baby for diseases you've never had. You've never had measles, polio, or diphtheria. How does your body know how to fight something it's never encountered?"*

Watch For

Ask students to name vaccines they've received. Then ask: How does a shot you got at 2 months old still protect you at 17? Collect hypotheses.

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling two connected systems: how your individual immune system builds memory, and how a vaccinated population creates collective protection."*

Watch For

Have students read objectives. Pre-teach 'memory B cell' and 'herd immunity threshold' with visual diagrams.

Slide 3: Big Question

3 min

Say: *"Here's what most people get wrong about vaccines: they're not just about you. Your vaccination protects people around you who CAN'T be vaccinated — babies, cancer patients, organ recipients. At what point does YOUR decision become THEIR fate?"*

Watch For

Quick-write: Students describe who they think is most affected when vaccination rates drop. Share out.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model that works at two scales — the molecular battle inside one person's body, and the epidemic math of an entire population."*

Watch For

Preview LEVER steps. Emphasize the shift from individual cells to population dynamics.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key players in this system? Some are choices we make, others are biological processes we can't directly control."*

Watch For

Guide sorting: Vaccination Rate and Antigen Exposure are external. Antibody Production, Memory Cells, Herd Immunity, and Transmission Rate are internal biological/epidemiological outcomes.

Slide 6: Activity 2: Connections

8 min

Say: *"Follow the chain: vaccine delivers antigen, antigen activates B cells, B cells make antibodies AND memory cells, memory cells provide long-term protection. Now zoom out: enough protected individuals create herd immunity that blocks transmission."*

Watch For

Students map connections at both scales. Key insight: individual immunity aggregates into population protection through a threshold effect.

Slide 7: Activity 3: Simulation

10 min

Say: *"First, let's watch one person's immune system learn. Then let's vaccinate a population and find the exact moment where disease transmission stops."*

Watch For

Run individual immune response simulation showing antibody peak and memory cell persistence. Then run herd immunity buildup, watching the sharp threshold transition.

Slide 8: Discoveries

5 min

Say: *"Your immune system is the most sophisticated learning machine on the planet. And when enough people's immune systems learn the same lesson, the whole community is protected — even those who can't learn it themselves."*

Watch For

Calculate herd immunity thresholds for different diseases. Discuss: What happens to a baby in a NICU when vaccination rates drop below 93% for measles?

Slide 9: STEM Challenge

12 min

Say: *"A new virus just emerged. R0 is 4.5. You have limited vaccine supply and 6 months to reach herd immunity. Who gets vaccinated first, and can you prove your plan works?"*

Watch For

Groups design vaccination rollout strategies with prioritization schedules, coverage targets, and model-based timeline predictions. Present and defend plans.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Vaccination Rate, Antigen Exposure
- **INTERNAL:** Antibody Production, Memory Cell Population, Herd Immunity Level, Disease Transmission Rate

Relationships

- **Antigen Exposure to Antibody Production:** POSITIVE (+) — Vaccine antigens activate B cells with matching receptors, triggering clonal expansion and differentiation into antibody-secreting plasma cells. More effective antigen presentation (through adjuvants or optimal vaccine design) produces a stronger initial antibody response.
- **Antibody Production to Memory Cell Population:** POSITIVE (+) — During the immune response, a fraction of activated B cells and T cells differentiate into long-lived memory cells rather than short-lived effector cells. A stronger initial immune response (more antibody production) generally produces a larger and more diverse memory cell population.
- **Vaccination Rate to Herd Immunity Level:** POSITIVE (+) WITH THRESHOLD — As vaccination rate increases, herd immunity builds gradually but the protective effect is nonlinear — disease transmission drops sharply when coverage crosses the herd immunity threshold (1 - 1/R0). Below the threshold, outbreaks can still occur; above it, transmission chains break.

Key Discoveries

- Vaccines work by training the immune system to recognize a pathogen's antigens WITHOUT causing disease — the adaptive immune system produces the same memory cells it would during a real infection, but without the danger
- Antibody levels decline after vaccination, but long-lived memory B cells and memory T cells persist for years or decades, ready to mount a rapid secondary response that is 10-100 times faster than the primary response
- Herd immunity is a threshold effect, not a gradual process — below the threshold, outbreaks can still occur; above it, transmission chains break and the disease cannot sustain spread through the population
- People who cannot be vaccinated (infants, immunocompromised patients, organ transplant recipients) depend entirely on herd immunity for protection — when vaccination rates drop below the threshold, these vulnerable individuals are the first to be harmed

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Vaccines inject the disease into your body and make you sick"	Vaccines do not contain active, disease-causing pathogens. Depending on the type, they contain weakened (attenuated) pathogens that cannot cause disease, killed pathogens, purified protein subunits, or mRNA instructions for making a single viral protein. These trigger an immune response — which can cause mild side effects like soreness, low fever, and fatigue as the immune system activates — but this is fundamentally different from having the disease. The side effects are signs that your immune system is learning.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"If vaccines work, why do vaccinated people sometimes still get sick?"	No vaccine is 100% effective — efficacy ranges from 40% (flu) to 97% (measles). Some vaccinated individuals don't develop full immunity due to individual variation in immune response. Additionally, immunity can wane over time, which is why boosters are recommended. However, vaccinated individuals who do become infected almost always experience milder illness, shorter duration, and dramatically lower risk of hospitalization or death compared to unvaccinated individuals. The model shows that even imperfect individual protection, when spread across a population, creates powerful herd immunity.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Herd immunity can be achieved through natural infection instead of vaccination"	While natural infection does produce immunity, achieving herd immunity through infection requires a massive number of people to actually get sick — and for diseases like measles, COVID-19, and influenza, this means widespread hospitalization, disability, and death. For measles (R0=15), 93% of the population would need to be infected, resulting in hundreds of thousands of deaths in a country like the US. Vaccination achieves the same herd immunity without the disease burden. The model clearly shows identical herd immunity curves whether immunity comes from vaccination or infection — but the human cost is orders of magnitude different.